Chapter 9

The main sequence

In this chapter we look at the inner workings of a star once it has settled into a main sequence existence. We start by looking at the sources of stellar energy, and then we look at the physical processes that govern stellar structure.

9.1 Stellar energy sources

When material collapses to form a star, there is gravitational potential energy stored from the (negative) work done by gravity in bringing the material together. We might wonder how long this energy supply will last. We start by calculating the gravitational potential energy in a uniform sphere.

9.1.1 Gravitational potential energy of a sphere

We are dealing with systems of large numbers of particles. Therefore, rather than thinking in terms of individual particles of mass m, we can think of a fluid of average density ρ. (The density is simply the mass per particle, multiplied by the number of particles per unit volume.) In this section, we will evaluate the potential energy for a uniform (constant density) sphere. Even though real objects might not be exactly uniform or spherical, the results will generally only change by numerical factors of order unity.

We begin by calculating the gravitational potential energy of a uniform sphere of mass M, radius R, and density ρ. These quantities are related by

$$M = (4\pi/3)\, R^3\rho \tag{9.1}$$

The gravitational potential energy is the work required to bring all of the material from far away (infinity) to the final configuration. The final result does not depend on the order in which the various parts of the sphere are assembled, so we do the calculation in the easiest way that we can envision. We can think of the sphere as being made up of shells (Fig. 9.1). We can assemble the sphere one shell at a time, starting with the smallest.

Let's assume that we have already assembled shells through radius r. We now want to calculate the work done to bring in the next shell. The thickness of the shell is $\mathrm{d}r$. The volume of the (thin) shell is its surface area multiplied by its thickness:

$$\mathrm{d}V = 4\pi r^2\,\mathrm{d}r \tag{9.2}$$

The mass contained in the shell is the volume multiplied by the density:

$$\mathrm{d}M = 4\pi r^2\rho\,\mathrm{d}r \tag{9.3}$$

The total mass of material already assembled is

$$M(r) = (4\pi/3)R^3\rho \tag{9.4}$$

The quantity $M(r)$ is important since the shell that ends up at radius r will only feel a net force from material inside it. Even after we bring in more material outside this shell, the net force exerted on any particle in the shell by any matter outside radius r is zero. Also, for mass in the shell, the mass $M(r)$ exerts a force equal to that which would be exerted by the same mass all located at the center of the sphere.

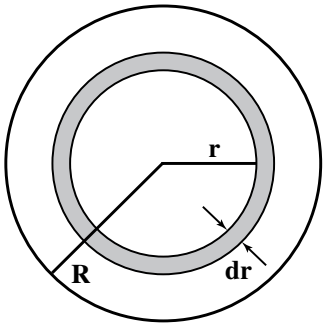

Fig 9.1. We model stars by studying spherical shells.

For any two point masses, remember the gravitational potential energy (relative to infinity) is given by

$$U = -Gm_1m_2/r \tag{9.5}$$

We let $m_1 = M(r)$ and m_2 equal the mass of the shell, dM. The work to bring this shell is

$$\begin{aligned} \mathrm{d}U(r) &= -GM(r)\,\mathrm{d}M/r \\ &= -G(4\pi/3)(4\pi)\rho^2 r^4\,\mathrm{d}r \end{aligned} \tag{9.6}$$

To find the effect of all the shells, we integrate the quantity dU from $r = 0$ to $r = R$:

$$\begin{aligned} U &= \int_0^R \mathrm{d}U(r) \\ &= -G(4\pi/3)(4\pi)\,\rho^2 \int_0^R r^4\,\mathrm{d}r \\ &= -G(4\pi/3)(4\pi)\,\rho^2\,(1/5)\,R^5 \qquad (9.7) \\ &= -(3/5)G[(4\pi/3)\,\rho\,R^3]^2/R \\ &= -\left(\frac{3}{5}\right)\frac{GM^2}{R} \end{aligned}$$

For other shaped objects, the gravitational potential energy is generally proportional to $-GM^2/R$, where R is some average length. The constant of proportionality is generally close to unity.

The thermal energy is $(3/2)kT$ per particle. The total thermal energy is then

$$K = (3/2)NkT \tag{9.8}$$

where N is the total number of particles in the cloud. If the mass per particle is m, and the total mass of the cloud is M, then

$$N = M/m \tag{9.9}$$

or

$$K = (3/2)(M/m)kT \tag{9.10}$$

9.1.2 Gravitational lifetime for a star

It is possible for stars to use their stored gravitational potential energy to power the star. In this section, we calculate how long this process can go on. For example, could the Sun be powered in this way even now? To see, we estimate the gravitational lifetime, t_g. This lifetime is the stored energy divided by the rate at which the energy is being lost. The rate at which the energy is being lost is the luminosity. That is

$$\begin{aligned} t_g &= \frac{E}{\mathrm{d}E/\mathrm{d}t} \\ &= \frac{E}{L} \end{aligned} \tag{9.11}$$

where L is the luminosity.

We estimate E as the negative of the Sun's current gravitational potential energy:

$$\begin{aligned} E &\cong \frac{(3/5)GM^2}{R} \\ &= \frac{(0.6)(6.67\times10^{-8}\ \mathrm{dyn\ cm^2/g^2})(2\times10^{33}\mathrm{g})^2}{(7\times10^{10}\ \mathrm{cm})} \\ &= 2\times10^{48}\ \mathrm{erg} \end{aligned} \tag{9.12}$$

This gives a lifetime, called the *Kelvin time*,

$$\begin{aligned} t_g &= \frac{(2\times10^{48}\ \mathrm{erg})}{(4\times10^{33}\ \mathrm{erg/s})} \\ &= 5\times10^{14}\ \mathrm{s} \\ &= 2\times10^{7}\ \mathrm{yr} \end{aligned}$$

This is a lifetime of 20 million years. It may sound like a long time. However, we know from geological evidence that the Earth has been around for over four billion years. This means that the Sun must be at least that old. Therefore, the Sun (and presumably other stars) cannot exist in a stable configuration on stored gravitational energy.

9.1.3 Other energy sources

An alternative source of energy could be chemical reactions. After all, we use chemical reactions to make automobiles go on Earth. We can estimate the amount of energy stored in the Sun, capable

of being released in chemical reactions. Typical energies of these chemical reactions should be equivalent to some fraction of the binding energy of molecules that might be formed and destroyed in these reactions. This means that we might expect something like 1 eV per atom in the Sun. The total chemical energy is then 1 eV multiplied by the total number of atoms in the Sun, M/m. The energy available is then

$$E = \frac{(1\ \text{eV})(1.6 \times 10^{-12}\ \text{erg/eV})(2 \times 10^{33}\ \text{g})}{(1.67 \times 10^{-24}\ \text{g})}$$

$$= 2 \times 10^{45}\ \text{erg}$$

This is much less than is stored in gravitational potential energy, so chemical reactions clearly cannot provide a longer term energy source for the Sun.

You might wonder if we have not been too casual in our dismissal of chemical reactions as a possible energy source for the Sun. After all, we didn't even say what chemical reactions might be involved. The important point is that *any* chemical reaction involves moving electrons from atom to atom, and the energies associated with this are a few eV, independent of what the reaction is. Suppose there were some very energetic reaction that produced 10 eV per atom, that would increase the chemical energy by a factor of ten but it would still be many orders of magnitude short of the required amount. If our estimate had shown that chemical reactions might work, then we would have to worry about the details, figure out what reactions were important and then do a more accurate calculation of the stored energy. However, our estimate tells us that it is not worth wasting our time on the details. These types of calculations, called *order of magnitude calculations*, are very important in astronomy. They help us eliminate processes that obviously don't work and allow us to focus our attention on possibilities that might.

The answer to the problem of stellar energy sources is nuclear reactions. The typical energies available in nuclear reactions are about 1 MeV per atom, instead of 1 eV. This is an improvement of a factor of 10^6. To see how nuclear reactions provide energy for the Sun, we look at some of the elements of nuclear physics.

9.2 Nuclear physics

9.2.1 Nuclear building blocks

We have already seen that the positive charge in atoms is confined to the small nucleus (10^{-13} cm across). The nucleus is composed of protons and neutrons. Because they are the building blocks of the nucleus, we call them *nucleons*. The proton has charge $+e$ (where $-e$ is the charge on the electron). Since atoms are neutral, the number of electrons orbiting the nucleus equals the number of protons in the nucleus. The chemical properties of an atom depend on the number of orbiting electrons. Therefore, the number of protons in the nucleus ultimately determines the identity of the element. We designate the number of protons by the symbol Z, called the *atomic number*. The charge on the nucleus is $+Ze$. The highest naturally occurring value of Z is 92 (uranium), with approximately a dozen "man-made" elements with higher values of Z.

Neutrons are electrically neutral. The mass of the neutron is slightly greater than the mass of the proton. Nuclei with the same numbers of protons can have different numbers of neutrons N. The total number of nucleons in the nucleus is called the *mass number*, $A = Z + N$. The mass of a given nucleus is approximately Am_p. Two nuclei with the same number of protons but different numbers of neutrons are called *isotopes* of the same element. We generally designate an element by a letter symbol (e.g. H for hydrogen), preceded by a superscript giving A, followed by a subscript giving Z. (This subscript is redundant since the symbol tells us Z, and is sometimes left out.) For example, the three known isotopes of hydrogen are $^1H_1$, $^2H_1$, $^3H_1$ or simply ^1H, ^2H, ^3H.

For any given number of protons, we do not find an arbitrary number of neutrons. In general, we find that the stable elements have approximately equal numbers of protons and neutrons. For larger values of Z the stable nuclei have a slightly larger number of neutrons than protons.

We now look at the force that holds nuclei together. We generally refer to four forces in nature. In order of decreasing strength, they are *strong nuclear, electromagnetic, weak nuclear* and *gravity*. The nuclear forces are shorter ranged, as

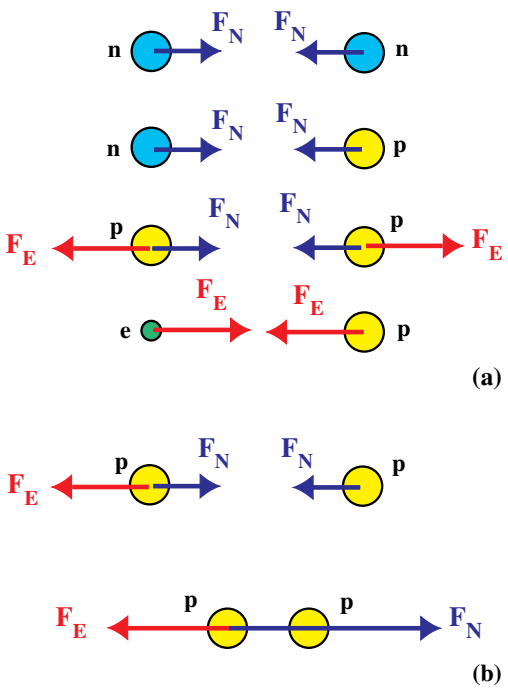

Fig 9.2. Properties of the nuclear and electrical forces between neutrons and protons. (a) The nuclear force, F_N, is the same for a proton and a proton, a neutron and a neutron, or a proton and a neutron. The electrical force only acts between the proton and the proton. The magnitude of the electrical force is the same as that between a proton and an electron. (b) How the forces vary with distance. As the two protons are brought closer together, the electric repulsion becomes stronger, but the nuclear attraction becomes stronger faster.

opposed to the electromagnetic and gravitational forces that can be felt at very large distances. For example, we look at the force between two protons. When they are far apart, the electric force dominates, and the protons repel each other. When they are close together, the attractive nuclear force dominates. The strong nuclear force between two protons is the same as between two neutrons and is the same as between a neutron and a proton (Fig. 9.2).

This gives us an idea of the role of a neutron in a nucleus. If we just have protons, the electric repulsion will be appreciable, and the nucleus will not be stable. If we add neutrons, we have additional binding from the nuclear force but no additional electrical repulsion. (The electric repulsion is actually reduced, since the neutrons keep the protons farther apart.) This explains why a nucleus needs approximately as many neutrons as protons.

9.2.2 Binding energy

Since the nuclear force is attractive, we must do work to move two nucleons apart. The work required to disassemble a nucleus is the *binding energy* of the nucleus (Fig. 9.3). The binding energy is analogous to the binding energy of an atom – the energy required to separate an electron from the rest of the atom, doing work against the electrical attraction. Since the nuclear force is so strong, nuclear binding energies are greater than atomic binding energies by a factor of about 10^6. We measure nuclear binding energies in MeV, rather than eV. The greater the binding energy of a nucleus, the more work we must do to get the nucleus apart. Therefore, a larger binding energy means a more stable nucleus.

The binding energies of nuclei are so large that we can measure them directly by comparing the mass of a nucleus with the masses of its components. This follows from Einstein's relationship between mass and energy ($E = mc^2$) and conservation of energy

$$M_{\text{nucleus}}c^2 + \text{BE} = Zm_pc^2 + Nm_nc^2 \tag{9.13}$$

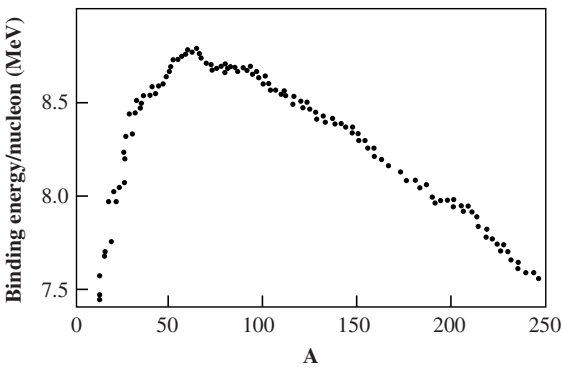

Fig 9.3. We must do work to break a nucleus apart. This work is the binding energy. Shown here are nuclear binding energies for nuclei that are found in nature. The horizontal axis is the mass number A, and the vertical axis the binding energy, divided by the number of nucleons in the nucleus (BE/A). Nuclei with higher binding energies per nucleon are more stable.

where BE is the binding energy. This means that the mass of a nucleus is less than the masses of its components by BE/c^2. This is also true for atoms. However, the binding energy for atoms is so small that the mass difference is negligible (about 10^{-6} of the nuclear binding energy).

Example 9.1 Proton rest energy
Compute the rest energy of a proton and express the result in MeV.

SOLUTION

$$E = (1.67 \times 10^{-24}\,\text{g})(3.0 \times 10^{10}\,\text{cm/s})^2$$
$$= 1.5 \times 10^{-3}\,\text{erg}$$
$$= \frac{1.5 \times 10^{-3}\,\text{erg}}{1.6 \times 10^{-12}\,\text{erg/eV}}$$
$$= 9.4 \times 10^{8}\,\text{eV}$$
$$= 940\,\text{MeV}$$

Example 9.2 Deuteron binding energy
Calculate the binding energy of the deuteron, an isotope of hydrogen containing one proton and one neutron, given the following data:

$$m_p = 1.6726 \times 10^{-24}\,\text{g}$$
$$m_n = 1.6749 \times 10^{-24}\,\text{g}$$
$$m_d = 3.3436 \times 10^{-24}\,\text{g}$$

SOLUTION
The binding energy is given by

$$BE = (m_p + m_n - m_d)c^2$$
$$= (3.9 \times 10^{-27}\,\text{g})(2.9979 \times 10^{10}\,\text{cm/s})^2$$
$$= 3.6 \times 10^{-6}\,\text{erg}$$
$$= 2.2\,\text{MeV}$$

By comparing the result of this example with the previous example, we see that the nuclear binding energy is a few tenths of a percent of the total rest energy. This mass difference is relatively easy to measure.

9.2.3 Nuclear reactions

We have seen that there are large amounts of energy involved in holding a nucleus together. However, we must still have a way of liberating that energy. If we hope to power stars by liberating nuclear energy, a nuclear reaction in which the products have a greater binding energy than the reactants must be found. Then, the difference in energy will be available to heat the surroundings. Before we discuss the particular nuclear reactions that work in stars, we look at the types of nuclear reactions that can take place.

The first type of reaction is a *decay*. In a decay, a nucleus emits a particle. Depending on the type of particle that is emitted, the result is either a different nucleus, or the same nucleus in a lower energy state. We identify three types of decay, depending on the type of particle that is emitted. The three types are called alpha (α), beta (β), and gamma (γ).

An *alpha particle* is the nucleus of the most common form of helium, ^{4}He. This means that an alpha particle has two protons and two neutrons. This particular combination is very stable and can be emitted as a single group. The final nucleus has two fewer protons and two fewer neutrons than the original nucleus. This means that the element changes as the atomic number Z decreases by two. The mass number A decreases by four. This process can only take place if the resulting nucleus is more stable than the original nucleus. This means that the binding energy of the final nucleus must be greater than that of the original nucleus. Once the alpha particle is far enough from the nucleus, it no longer feels the nuclear attraction. However, since it has a charge of $+2e$, it feels the Coulomb repulsion of the remaining nucleus. The alpha particle accelerates away from the nucleus. The energy liberated in the reaction is carried away in the form of the kinetic energy of the alpha particle.

A *gamma-ray* is simply a high energy photon. Since it carries away neither mass nor charge, neither A nor Z change. The identity of the nucleus does not change in gamma decay. The photon does carry away energy. The protons and neutrons in a nucleus must also be treated quantum mechanically as being waves. This means that there are allowed states for the motions of the nucleons within the nucleus. A gamma-ray is emitted when a transition is made from a higher energy state to a lower energy state. The energy of the gamma-ray is equal to the energy difference

between the states. We can learn a lot about the internal structure of nuclei by studying the energies of gamma-rays that are emitted, just as optical spectroscopy tells us about the structure of atoms.

A *beta particle* is an electron or a *positron*. A positron is the *antiparticle* to the electron. Every particle has a corresponding antiparticle. A particle and its antiparticle have identical masses, but all other properties, such as charge, are the negative of each other. When a particle and an antiparticle come together, they can convert all of their mass into energy – in other words they annihilate each other – without violating any conservation laws. In a beta decay, an electron or a positron is emitted. The charge of the nucleus increases by $+e$ (or $-e$ if a positron is emitted), but the mass number A remains unchanged. The net result is to change a neutron into a proton. (The reverse, changing a proton into a neutron is also possible, but requires a source of energy, because the neutron is more massive than the proton.) When a neutron changes to a proton, Z increases by one, but N deceases by one, again leaving A unchanged.

In beta decays, an additional particle, called a *neutrino*, is emitted. Neutrinos were originally postulated because an analysis of beta decays indicated that some energy and angular momentum were being lost in the process. It was therefore assumed that a massless neutral particle was carrying away this energy and angular momentum. If neutrinos are truly massless (and we will have more to say on this point in Part VI of this book), then they travel at the speed of light. The existence of the neutrino was verified experimentally in the 1950s. Neutrinos do not interact with matter via the strong nuclear or electromagnetic forces. This means that a reaction in which a neutrino is involved must proceed by the weak nuclear force. The weak force is so weak that neutrinos rarely interact with matter. Weak decays also proceed at much slower rates than do strong decays.

The basic beta decay reaction is

$$\mathrm{n} \rightarrow \mathrm{p} + \mathrm{e}^- + \bar{\nu} \tag{9.14}$$

The bar over the ν indicates an antineutrino. In free space, the average time for this reaction to take place is 11 min. In nuclei, the reaction can take place only if the resulting proton ends up in a lower energy state than the original neutron. This places an upper limit on the number of neutrons that a nucleus with a given Z can have and still be stable. If we have too many neutrons, some will beta decay and become protons in lower energy states.

Another type of nuclear reaction is called *fission*, in which a nucleus breaks into smaller parts. The controlled fission of uranium provides the energy in current nuclear power plants. Generally, in fission, a very heavy nucleus breaks into a form that is of more stable, middle mass nuclei. From Fig. 9.3, we see that the most stable nuclei are the ones with intermediate masses, with iron being the most stable. The fission of uranium or plutonium has been used in "atomic bombs".

The final type of nuclear reaction we will consider is *fusion*, in which lighter nuclei can come together to build heavy nuclei. Fusion is important in stellar energy generation, because there is an ample supply of lighter elements. One problem with fusion is that it is hard to start. When two nuclei are far apart the nuclear force between them vanishes. They feel only the electrical repulsion. To bring nuclei close enough together for the nuclear force to take over, we must do work against the electrical repulsion. This requires accelerating the particles to high energies and letting them collide. Since they have high energies, they will be slowed by the electrical repulsion, but not stopped. On the Earth, we can accomplish this acceleration for small quantities of matter in particle accelerators, but this is impractical for large quantities of material. This is the major problem that must be overcome before we can realistically think about using fusion as an economical source of energy on the Earth.

9.2.4 Overcoming the fusion barrier

One way to increase the energies of particles is to raise their temperature. If the temperature is high enough, particles will be moving fast enough to overcome the electrical repulsion, and fusion reactions can take place. On the Earth this poses problems, since we have trouble containing a gas at the required temperature, tens of millions of

kelvin. However, in stars the material is confined by gravity. If sufficiently high temperatures are reached, the nuclear reactions can take place. If the temperatures are not high enough to support nuclear reactions, then we don't have a star.

We can estimate the temperature required for nuclear fusion to take place. Let's consider the case of two protons. Their electric potential energy when they are a distance r apart is

$$U = e^2/r \tag{9.15}$$

Suppose we start with two protons far apart, so we can take their potential energy to be zero. We let K be the kinetic energy of the particles at this point. We would like the particles to come to rest (zero kinetic energy) a distance r_p, the radius of the proton, apart. From conservation of energy, the kinetic energy when the protons are far apart must be

$$K = e^2/r_p$$
$$= \frac{(4.8 \times 10^{-10}\ \text{esu})^2}{(1 \times 10^{-13}\ \text{cm})}$$
$$= 2 \times 10^{-6}\ \text{erg}$$

If we divide this by the Boltzmann constant, k, we have an estimate of the temperature at which the average kinetic energy of the particles in the gas is equal to this energy. This gives a temperature of 2×10^{10} K, a very high temperature. If we had considered the case of nuclei with charges $+Z_1e$ and $+Z_2e$, the potential energy becomes $Z_1Z_2e^2/r$. This means that even higher temperatures are needed for the fusion of higher charged nuclei.

Actually, the temperature doesn't have to be as high as our calculation suggests. If the temperature is as high as we calculate, then the energy of the *average* particle will be high enough for it to participate in fusion. We have already seen that not all particles in a gas have the average energy. Even at lower temperatures there will be some particles with a high enough energy to undergo fusion. For particles with a Maxwell–Boltzmann velocity distribution, the probability of finding a particle with an energy between E and $E + dE$ is proportional to a term like the $\exp(-E/kT)$ in the Boltzmann equation. That is

$$P(E)dE \propto E^{1/2}\, e^{-E/kT}\, dE \tag{9.16}$$

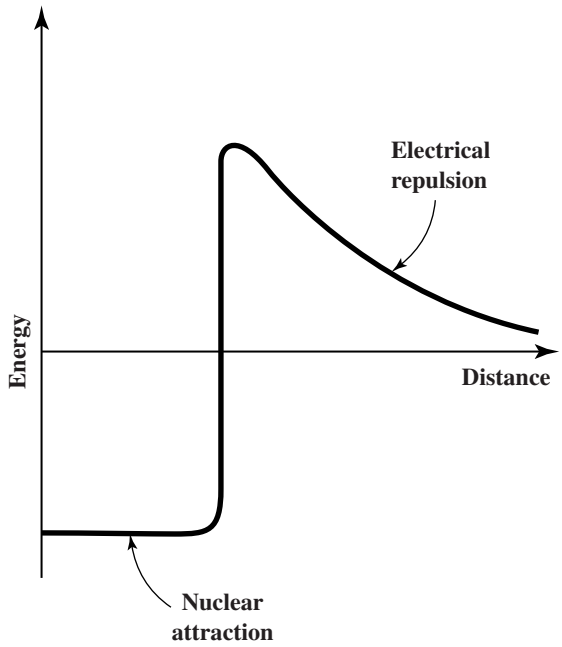

Fig 9.4. The potential energy for two protons as a function of distance. This includes the electrical force and an estimate of the nuclear force.

There is another effect which allows nuclear reactions to take place at a lower temperature than we have calculated. Fig. 9.4 shows the potential energy of two protons as a function of distance between them. For the most part, the force is repulsive (electrical), but if the protons are brought close enough together the force would be attractive. Suppose E_0 is the minimum energy for a particle to be able to overcome the Coulomb repulsion. We can see that this energy E_0 is the "height" of the potential energy barrier. Particles with energies greater than E_0 will pass through the barrier. However, according to classical physics, particles with energies less than E_0 will reach a point at which their energy equals the height of the barrier at some distance r_0. At this point all of the energy is in the form of potential, so the kinetic energy is zero. The particle has stopped and is about to head back in the other direction. The point at which the particle turns back is called the *turning point*.

Quantum mechanically, we should talk about the probability of finding a particle in various places. Let's look again at the particle with energy

less than E_0. Initially, the particle has a high probability of being found farther from the proton than the classical turning point. As the classical turning point is approached, the probability cannot suddenly go from some finite value to zero, because that probability is related to a wave phenomenon. So, as we go into the barrier, the probability of finding the particle falls off gradually. This means that there is some probability of finding the particles closer together than the classical turning point. This phenomenon is called *barrier penetration* or *tunneling*.

In general, a particle can penetrate a distance approximately equal to its wavelength, h/mv. More precisely, the probability of penetrating a distance x is related to the wavelength by

$$P(x)\mathrm{d}x \propto \mathrm{e}^{-ax/\lambda}\,\mathrm{d}x$$
$$= \mathrm{e}^{-axmv/h}\,\mathrm{d}x \qquad (9.17)$$

where a is some constant. Suppose we have to tunnel a distance x, equal to the classical turning point r_0, defined by the initial kinetic energy being equal to the electrostatic potential energy at r_0, or

$$\frac{1}{2}mv^2 = \frac{Z_1 Z_2 e^2}{r_0}$$

Substituting into equation (9.17), we have the probability of penetrating r_0:

$$P(r_0) \propto \mathrm{e}^{-aZ_1Z_2e^2/2hv}$$
$$\propto \mathrm{e}^{-b/E^{1/2}}$$

where b is a constant. The effective area of the nucleus for a reaction is approximately λ^2 and is proportional to $1/E$. When we combine this with the Maxwell–Boltzmann velocity distribution, we find the probability of a reaction by nuclei of energy E is proportional to

$$\mathrm{e}^{(-E/kT - b/E^{1/2})}$$

This means that at a given T there will be a most likely value of E for reactions to take place. This is known as the *Gamow peak* shown in Fig. 9.5. It turns out that most reactions involve particles that are on the low energy side of the velocity distribution. These lower energy particles have longer wavelengths and can penetrate farther into the barrier. The exponential behavior of the reaction rates also makes them very sensitive to temperature. That is, small changes in T will produce large changes in the reaction rates.

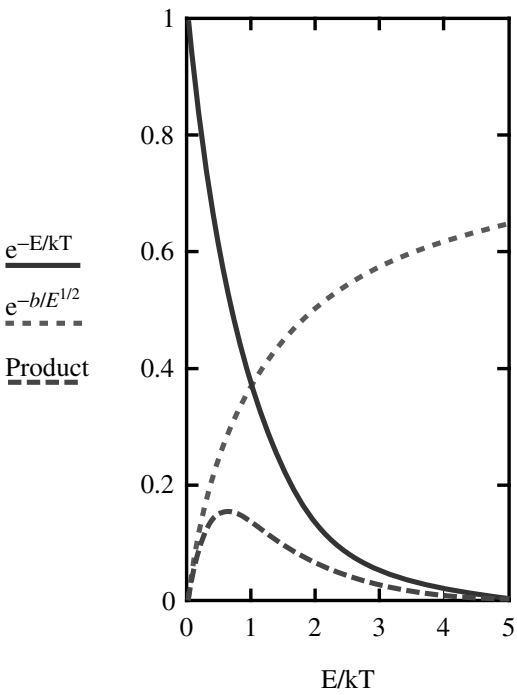

Fig 9.5. The probability of a nuclear fusion, as a function of the particle energy E, at a given gas temperature T. This shows the combined effects of the location of the classical turning point and the quantum-mechanical tunneling.

9.3 Nuclear energy for stars

When a star is on the main sequence, its basic source of energy is the conversion of hydrogen into helium. We start with four protons and end up with one ^{4}He nucleus. However, it is unlikely that four protons will get close enough to directly form a ^{4}He nucleus in a single reaction. There are different series of reactions that achieve this net result, and they will be discussed below.

We can calculate the energy released by converting four protons to one ^{4}He by comparing their masses. We find that

$$4m_\mathrm{p} - m(^4\mathrm{He}) = 0.007(4m_\mathrm{p}) \qquad (9.18)$$

This means that 0.007 of the mass of each proton is converted into energy. (The 007 will be familiar to James Bond fans.)

Example 9.3 Lifetime of the Sun
Estimate the lifetime of the Sun for producing energy at its current rate from nuclear fusion.

SOLUTION
If 0.007 of the mass of each proton in the Sun is converted into energy, and if we assume that most of the mass of the Sun was originally in the form of protons, then 0.007 of the Sun's total mass is available for conversion into energy. The total energy available is therefore

$$E = 0.007\, M_\odot\, c^2$$
$$= (0.007)(2.0 \times 10^{33}\ \text{g})(3.0 \times 10^{10}\ \text{cm/s})^2$$
$$= 1.3 \times 10^{52}\ \text{erg}$$

The lifetime is this energy divided by the luminosity:

$$t_n = E/L$$
$$= \frac{(1.3 \times 10^{52}\ \text{erg})}{(4 \times 10^{33}\ \text{erg/s})}$$
$$= 3.2 \times 10^{18}\ \text{s}$$
$$= 1 \times 10^{11}\ \text{yr}$$

However, we think that only 10% of the mass of the Sun is in a region hot enough for nuclear reactions – the core, so we must lower our estimate by a factor of ten. This leaves us with a lifetime of ten billion years. We think that the Sun has already lived half of this time.

We now look at actual nuclear reactions, the net result of which is to convert four protons in a ^{4}He nucleus, plus energy. The basic series of reactions in stars like the Sun is called the *proton–proton chain*, because it starts with the direct combination of two protons:

$$\text{p} + \text{p} \rightarrow \text{d} + \text{e}^+ + \nu \tag{9.19}$$

which we can also write as

$$^1\text{H} + {}^1\text{H} \rightarrow {}^2\text{H} + \text{e}^+ + \nu$$

This process requires a temperature of 10^7 K. In this reaction the e^+ is a positron, the antiparticle to the electron. We can see by the presence of the neutrino that this process is a weak interaction, and therefore goes very slowly.

Once the deuteron has been created, it can quickly react with another proton:

$$\text{d} + \text{p} \rightarrow {}^3\text{He} + \gamma \tag{9.20}$$

which we can also write as

$$^2\text{H} + {}^1\text{H} \rightarrow {}^3\text{He} + \gamma$$

If each of these reactions takes place twice, then we have started with six protons, and we now have two ^{3}He nuclei. (^{3}He is an isotope of helium with two protons and one neutron. It is one of the few stable nuclei with fewer neutrons than protons.) The ^{3}He nuclei combine to give

$$^3\text{He} + {}^3\text{He} \rightarrow {}^4\text{He} + {}^1\text{H} + {}^1\text{H} \tag{9.21}$$

Note that we have two of the protons back. The net result is that we have converted four protons into a ^{4}He nucleus, along with two positrons, two gamma-rays and two neutrinos. The energy given off in this chain is carried away by the positrons, gamma-rays and neutrinos. The positrons will immediately scatter off (or annihilate) particles in the gas, resulting in heating of the gas. The gamma-ray will travel a small distance before being absorbed, also heating the gas. The hot gas will emit new photons, which are absorbed. This process of absorption and re-emission of the photons takes place until photons are close enough to the surface to escape. The neutrino interacts so weakly that it will escape from the star completely.

Stars more massive than the Sun have higher central temperatures than the Sun. This means reactions involving higher charged nuclei are possible. One such process is responsible for the buildup of elements heavier than helium. It is called the *triple-alpha process*, because the net result is to convert three alpha particles into a ^{12}C nucleus. The repulsion between two alpha particles is four times that between two protons, so higher temperatures, about 10^8 K, are needed. The first step in the chain is

$$^4\text{He} + {}^4\text{He} \rightarrow {}^8\text{Be} + \gamma \tag{9.22a}$$

We should note that the binding energy of two ^{4}He nuclei exceeds that of the ^{8}Be because the ^{4}He is such a stable nucleus. This tells us that the ^{8}Be should be unstable and break up. However, there will be so many alpha particles around that some of the ^{8}Be nuclei will capture an alpha

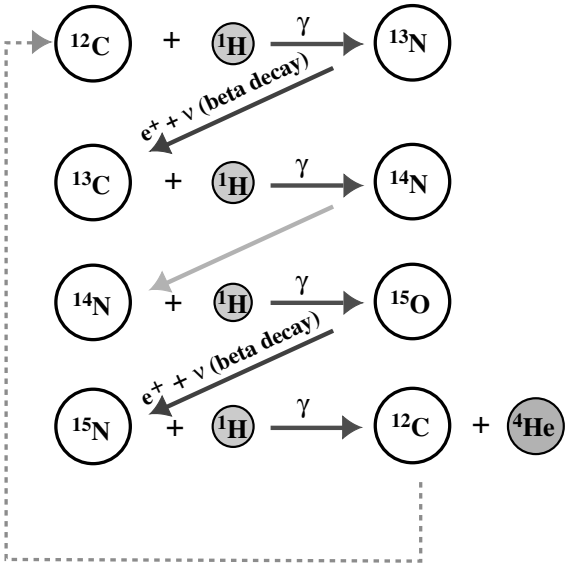

Fig 9.6. The CNO cycle. Solid arrows represent reactions. Symbols over these arrows indicate emitted particles. Dashed lines indicate when a created particle participates in another reaction.

particle before breaking up. If this did not happen, the buildup of heavier elements would be blocked. The combination of the ^{4}He and ^{8}Be gives

$$^4\text{He} + {}^8\text{Be} \rightarrow {}^{12}\text{C} + \gamma \qquad (9.22b)$$

The triple-alpha process is also important as solar mass stars age and leave the main sequence (discussed in Chapter 10).

In massive stars, there is another scheme that is important in converting four protons into one ^{4}He nucleus. It is called the *CNO cycle*. The cycle is indicated graphically in Fig. 9.6, and the steps are:

$$^{12}\text{C} + {}^1\text{H} \rightarrow {}^{13}\text{N} + \gamma \qquad \text{(strong force)} \qquad (9.23)$$

$$^{13}\text{N} \rightarrow {}^{13}\text{C} + e^+ + \nu \qquad \text{(beta decay)} \qquad (9.24)$$

$$^{13}\text{C} + {}^1\text{H} \rightarrow {}^{14}\text{N} + \gamma \qquad \text{(strong force)} \qquad (9.25)$$

$$^{14}\text{N} + {}^1\text{H} \rightarrow {}^{15}\text{O} + \gamma \qquad \text{(strong force)} \qquad (9.26)$$

$$^{15}\text{O} \rightarrow {}^{15}\text{N} + e^+ + \nu \qquad \text{(beta decay)} \qquad (9.27)$$

$$^{15}\text{N} + {}^1\text{H} \rightarrow {}^{12}\text{C} + {}^4\text{He} \qquad \text{(strong force)} \qquad (9.28)$$

We see that the net result is the conversion of four protons into one ^{4}He nucleus plus two positrons, two neutrinos, and three photons. All nuclei created as intermediate products are used in the next step. In addition, the last step returns the ^{12}C we need to start the cycle, so the cycle can go again. In a sense, we can think of the ^{12}C as a catalyst for this cycle. (A catalyst helps something happen, but is not itself changed in the process.) There are more complicated versions of this cycle involving even heavier elements, but the basic ideas are the same.

As we try to build up heavier and heavier elements through fusion, the electrical repulsion becomes stronger. This becomes an effective barrier to the formation of heavier elements. However, no matter how high the Z a nucleus has, we can always get a neutron near it with no electrical repulsion. If the neutron is moving slowly, it can be captured by the nucleus. This can be important in stars, because some reactions provide free neutrons, so there are generally some free neutrons available. We can schematically represent what happens when the nucleus (Z, A) captures a neutron:

$$(Z, A) + \text{n} \rightarrow (Z, A + 1) \qquad (9.29)$$

What happens next depends on whether the rate of neutron capture is slow or rapid, compared to the rate of beta decay. If neutron capture is slow, we call the sequence of reactions an *s-process* (Fig. 9.7). In this situation the new nucleus

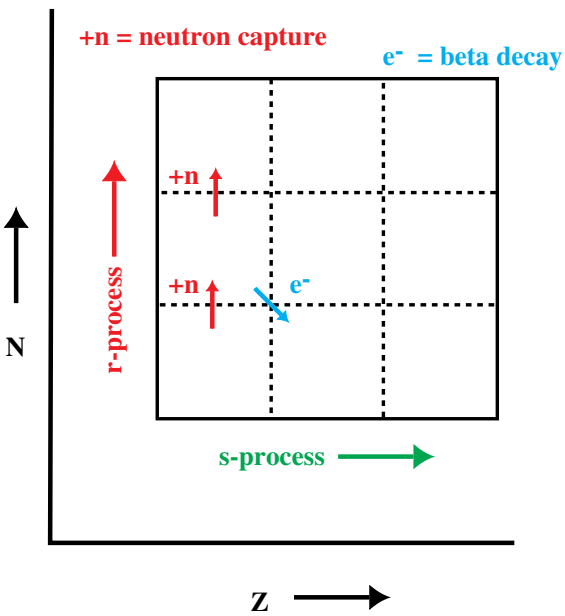

Fig 9.7 The r- and s-processes. The horizontal axis indicates increasing proton number Z, and the vertical axis indicates increasing neutron number N. In an r-process, a neutron is captured, and in an s-process the capture is followed by a beta decay. The +n next to an arrow indicates neutron capture. The e^- next to an arrow indicates a beta decay.

$(Z, A + 1)$ will beta decay before it can capture another neutron:

$$(Z, A + 1) \rightarrow (Z + 1, A) + e^- + \bar{\nu} \tag{9.30}$$

If the neutron capture is rapid we call the sequence of reactions an *r-process* (Fig. 9.7). The nucleus $(Z, A+1)$ will capture another neutron before it beta decays:

$$(Z, A + 1) + n \rightarrow (Z, A + 2) \tag{9.31}$$

In either case, the resulting nucleus can either beta decay or capture a neutron, depending on the relative rates. When we have a string of nuclei for which neutron capture is favorable, the r-process allows the buildup of neutron rich nuclei. This will go on until so many neutrons are added that a beta decay breaks the chain. The r- and s- processes can explain the abundances of many of the heavier nuclei. (It should be noted that these are not equilibrium processes.)

The various nuclear processes that we have discussed are responsible for the presence of the heavy elements around us. We will see in later chapters how this material is spread into interstellar space. The net result is to produce the abundances shown in Fig. 9.8. Nuclear physics determines which elements are the most abundant.

When we discuss stellar structure in the next section, we will treat the nuclear physics as something that is known. We assume that once we know the composition of some region and the temperature, we can specify which nuclear reactions are important. Moreover, we assume that we know how the reaction rates depend on temperature. This is a very important point. We have already seen in this section that tunneling by those nuclei with higher than average energy leads to a strong temperature dependence on the reaction rates. For example, the rates of some important reactions depend on temperature as T^7 or even higher powers.

We say that a collapsing object makes a transition from protostar to star when its primary source of energy generation is fusion via the p–p chain, rather than gravitational collapse. The changeover is not a sudden one, as the material closer to the center heats faster. Eventually, enough energy is generated internally for the collapse to halt. When this happens, the star reaches a stable condition. It is on the main sequence.

As nuclear reactions take place in the star, the composition of the star is actually changing. This change could affect the spectral type and luminosity of the star, while the basic structure doesn't change very much. These changes result in a main sequence that is a band on the HR diagram rather than a thin line. However, there is a line that we can identify as connecting the points on the HR diagram where stars of each spectral type first appear on the main sequence. We call this line the *zero-age main sequence*, or *ZAMS*. This line is shown relative to the main sequence band in Fig. 9.9.

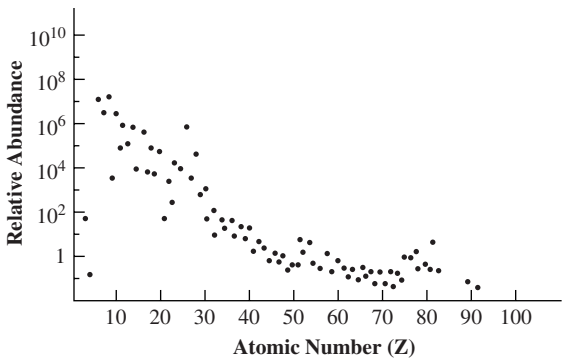

Fig 9.8 Cosmic abundances of the elements as a function of atomic number Z.

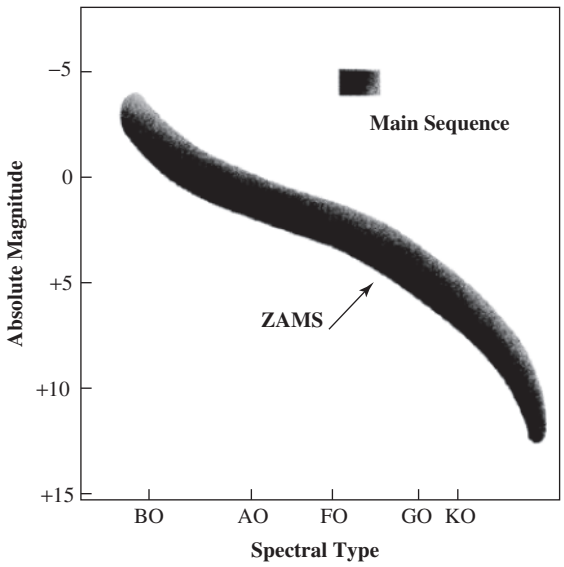

Fig 9.9 Zero-age main sequence. The main sequence on this HR diagram appears as a band, since stars on the main sequence become slightly brighter as their composition changes. The lower edge of this band represents the points where stars first appear on the main sequence, the ZAMS.

9.4 Stellar structure

The basic philosophy of stellar structure studies is that stars obey the laws of physics, so we should be able to predict and explain their structure by applying those laws. To do this, we must identify the basic physical processes that are important in stars, such as nuclear physics for energy generation. We must also be able to perform large numbers of intricate calculations. This latter facility is provided by modern computers.

Once we carry out stellar structure calculations, we can compare the predictions of the theories with observations. For example, if we put $1\,M_\odot$ of material into model calculations, we should come out with a star whose radius, temperature and luminosity match those of the Sun. If we put in different amounts of material, our calculations should reproduce the main sequence. We should also find stars in the same mass range as those on the main sequence. Stellar models should allow us to predict stellar evolution. They should also be able to tell us how changes in composition lead to changes in structure.

Stars are easier to analyze than some other astronomical objects because they have simple shapes. They are spheres. We also assume that their structure is spherically symmetric. That is, the conditions – temperature, density and composition – only depend on distance from the center, not the direction in which you are going away from the center. We can rotate the star through any angle about any axis through the center and not change the result. (This is not strictly true if the star is rotating or has a strong magnetic field.) To study a star we divide it into spherical shells, each of thickness dr, as shown in Fig. 9.1. If the density a distance r from the center is $\rho(r)$, then the mass contained in a shell of radius r and thickness dr is

$$\mathrm{d}M = \rho(r)\,\mathrm{d}V \tag{9.32a}$$

where dV ($= 4\pi r^2\,\mathrm{d}r$) is the volume of the shell. This gives

$$\mathrm{d}M = 4\pi r^2\,\rho(r)\,\mathrm{d}r \tag{9.32b}$$

Dividing both sides by dr gives us the rate at which we add mass as we go farther out from the center of the star:

$$\mathrm{d}M/\mathrm{d}r = 4\pi r^2\,\rho(r) \tag{9.33}$$

This condition is called *mass continuity*, and simply tells us how the rate of change of $M(r)$, mass interior to r, is related to the density at r. $M(r)$ is important because, for a spherical object, the gravitational force on an object a distance r from the center only depends on $M(r)$.

9.4.1 Hydrostatic equilibrium

The material in a shell of radius r is pulled toward the center of the star by the gravitational attraction of all the mass interior to that shell. Something must support the matter in the shell or else the star will collapse. That something is the pressure difference between the bottom of the shell and the top of the shell. This condition is called *hydrostatic equilibrium* (Fig. 9.10). Hydrostatic equilibrium applies to the Earth's atmosphere as well as in the oceans and in a glass of water. The weight of each layer of the fluid is supported by the pressure difference between the bottom and the top.

We can see how much of a pressure difference is needed by considering a small cylinder, of height dr and area dA, as shown in Fig. 9.10. The mass of the cylindrical element is

$$\mathrm{d}m = \rho(r)\,\mathrm{d}r\,\mathrm{d}A \tag{9.34}$$

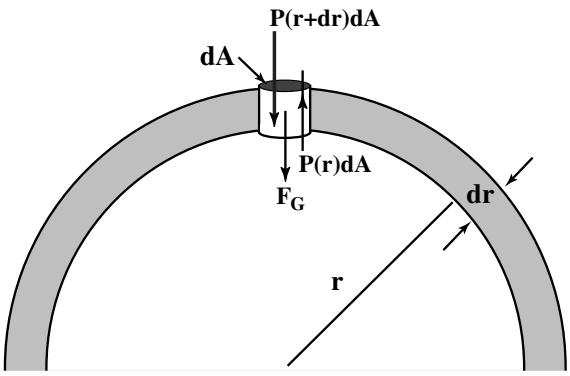

Fig 9.10 Hydrostatic equilibrium. The distance from the center of the star is r, and the thickness of the shell is dr. The density in the shell is $\rho(r)$. We consider the forces on a cylinder of height dr and end area dA. The pressure at r is $P(r)$.

The gravitational force depends on $M(r)$. In any particular model, we can find $M(r)$ by integrating equation (9.33):

$$M(r) = 4\pi \int_0^r \rho(r')r'^2\,\mathrm{d}r' \tag{9.35}$$

The gravitational force is given by

$$F_G = -GM(r)\,\mathrm{d}m/r^2 \tag{9.36}$$

We use the minus sign (−) to indicate that the force is directed downward. Taking dm from equation (9.34) gives

$$F_G = -[GM(r)/r^2]\rho(r)\,\mathrm{d}r\,\mathrm{d}A \tag{9.37}$$

We now look at the force exerted on the top and bottom of the cylinder by the pressure of the fluid. The difference between the upward force on the bottom and the downward force on the top is called the *buoyant force*, F_B. If $P(r)$ is the pressure at the bottom of the cylinder and $P(r + \mathrm{d}r)$ is the pressure at the top, then

$$\begin{aligned} F_B &= P(r)\,\mathrm{d}A - P(r + \mathrm{d}r)\,\mathrm{d}A \\ &= [P(r) - P(r + \mathrm{d}r)]\,\mathrm{d}A \end{aligned} \tag{9.38}$$

We have used the fact that force is pressure × area. We define the pressure difference between the top and the bottom, dP, as

$$\mathrm{d}P = P(r + \mathrm{d}r) - P(r) \tag{9.39}$$

We see that the pressure will decrease as r increases (since the pressure at the bottom of the shell must be greater than the pressure at the top), so dP is a negative number. Equation (9.38) then becomes

$$\begin{aligned} F_B &= \{P(r) - [P(r) + \mathrm{d}P]\}\,\mathrm{d}A \\ &= -\mathrm{d}P\,\mathrm{d}A \end{aligned} \tag{9.40}$$

The condition for hydrostatic equilibrium is

$$F_G + F_B = 0 \tag{9.41}$$

Substituting from equations (9.37) and (9.40) gives

$$-[GM(r)/r^2]\,\rho(r)\,\mathrm{d}r\,\mathrm{d}A - \mathrm{d}P\,\mathrm{d}A = 0 \tag{9.42}$$

We are interested in the rate at which the pressure changes with radius dP/dr, so we divide both sides by dA, add dP to both sides, and then divide both sides by dr to give

$$\mathrm{d}P/\mathrm{d}r = -[GM(r)/r^2]\,\rho(r) \tag{9.43}$$

This is sometimes called the *equation of hydrostatic equilibrium*. We can rewrite it more simply by noting that the quantity $GM(r)/r^2$ is equal to the local acceleration of gravity $g(r)$, so

$$\mathrm{d}P/\mathrm{d}r = -\rho(r)\,g(r) \tag{9.44}$$

The equation of hydrostatic equilibrium tells us that the denser the fluid is, the more rapidly P changes with r. This is because a denser fluid means a higher mass shell, and a stronger gravitational force pulling it in. This requires a larger pressure difference between the top and the bottom to support it. Also, the greater $g(r)$ is, the greater the gravitational force is pulling the shell in. This means that a larger $g(r)$ also requires a faster rate of change in the pressure.

Example 9.4 Central pressure of the Sun

Use the equation of hydrostatic equilibrium to estimate the central pressure of the Sun by considering the whole Sun as one shell.

SOLUTION

If we consider a whole star to be one shell, then dR = R, the radius of the star, and dP=P_C , the central pressure (taking the pressure at the surface to be zero). The equation of hydrostatic equilibrium then gives

$$P_C = (GM/R^2)\,\rho\,R$$

where ρ is the average density, and is approximately M/R^3, so

$$\begin{aligned} P_C &\cong (GM/R)(M/R^3) \\ &= GM^2/R^4 \end{aligned}$$

Substituting for the Sun gives

$$\begin{aligned} P_C &= \frac{(6.67 \times 10^{-8}\ \mathrm{dyn\ cm^2/g^2})(2 \times 10^{33}\ \mathrm{g})^2}{(7 \times 10^{10}\ \mathrm{cm})^4} \\ &= 1 \times 10^{16}\ \mathrm{dyn/cm^2} \end{aligned}$$

The actual value, obtained from stellar models, is about 20 times this value

Another equation that we use in modeling stars is the *equation of state*. The state of a gas is

described by the pressure, density and temperature. The equation of state relates those three quantities. We can write it in the general form

$$P = f(\rho, T) \tag{9.45}$$

The actual form of the function depends on the nature of the gas. For an ideal gas, the equation of state has the simple form

$$P = (\rho/m)\, kT \tag{9.46}$$

where m is the mass per particle. For a gas of relativistic particles, the equation of state is different, and we will discuss it in Chapter 10.

Example 9.5 Central temperature of the Sun
Use the result of the previous example and the equation of state for an ideal gas to estimate the central temperature of the Sun.

SOLUTION
To use the ideal gas law, we need to estimate the density. We simply use the average density, which is the mass divided by the volume. Since the hydrogen is completely ionized, there are an equal number of electrons and protons, so the average mass per particle is $(1/2)m_p$.

$$T = \frac{mP}{\rho k}$$

$$= \frac{mP(4\pi/3)\, R_\odot^3}{M_\odot\, k}$$

$$= \frac{(0.5)(1.67 \times 10^{-24}\,\mathrm{g})(1 \times 10^{16}\,\mathrm{dyn/cm^2})(4\pi/3)(7 \times 10^{10}\,\mathrm{cm})^3}{(2 \times 10^{33}\,\mathrm{g})(1.38 \times 10^{-16}\,\mathrm{erg/K})}$$

$$= 4.4 \times 10^7\,\mathrm{K}$$

This is a little larger than the model results, with $T \sim 1.5 \times 10^7$ K.

9.4.2 Energy transport

In making a stellar model we must also consider how energy gets from the inside of the star to the outside. In general, energy can be transported by conduction, convection and radiation. In stellar interiors conduction is not generally important. The energy transport must be such that the temperature $T(r)$ does not change with time. If it did, the star would not be stable for the multibillion year lifetime for stars like the Sun. For the temperature distribution to be constant with time, the energy entering any shell per second must equal the energy leaving that shell per second.

If radiation transport dominates, we can calculate the required temperature distribution, $T(r)$. We let $f(r)$ be the flux of radiation through a surface at radius r. If the surface emits like a blackbody, then

$$f(r) = \sigma\, T(r)^4 \tag{9.47}$$

We can find the rate at which $f(r)$ changes with $T(r)$,

$$\mathrm{d}f/\mathrm{d}T = 4\sigma\, T(r)^3 \tag{9.48}$$

If we interpret $\mathrm{d}f$ as the small change in f due to a small change in T, $\mathrm{d}T$, then

$$\mathrm{d}f = 4\sigma\, T(r)^3\, \mathrm{d}T \tag{9.49}$$

However, the change in flux passing through a given layer of the star must depend on the ability of the layer to absorb radiation. In Section 6.2 we saw that this is given by the absorption coefficient κ. In stellar structure it is more convenient to deal with the absorption coefficient per density of material. We therefore let $\kappa'(r)$ be the opacity per unit mass at r. This means that $\kappa'(r)\,\rho(r)$ gives the fraction of radiation absorbed per centimeter. Using these definitions

$$\mathrm{d}f = -\kappa'(r)\rho(r)f(r)\mathrm{d}r \tag{9.50}$$

The minus sign (−) tells us that the flux is decreased by the absorption.

We define the luminosity of a given layer as the flux f multiplied by the surface area of the layer,

$$L(r) = 4\pi r^2 f(r) \tag{9.51}$$

If we use equations (9.49) and (9.50) to eliminate $\mathrm{d}f$ and equation (9.51) to eliminate $f(r)$, we have

$$4\sigma T(r)^3\, \mathrm{d}T = -\frac{\kappa'(r)\,\rho(r)\,L(r)\,\mathrm{d}r}{4\pi r^2} \tag{9.52}$$

Solving for the luminosity gives

$$L(r) = -\left[\frac{16\pi r^2 \sigma T(r)^3}{\kappa'(r)\,\rho(r)}\right]\frac{\mathrm{d}T}{\mathrm{d}r} \tag{9.53}$$

This tells us how the rate of energy flow depends on the rate at which the temperature changes with r. In general, dT/dr is negative (the temperature drops with distance from the center), so the luminosity is positive. A more exact calculation gives essentially the same result as equation (9.53), but with the 16 replaced by 64/3.

From equation (9.53) we can see that the opacity per unit mass $\kappa'(r)$ is very important in determining the energy transfer, and therefore the structure of the star. The opacity depends on the composition of the star. Accurate stellar structure calculations require good knowledge of the opacity as a function of composition and temperature.

In addition to energy transport, we must also consider energy generation. If energy is generated in a particular shell, then the energy leaving that shell will exceed the energy entering the shell by the amount of energy generated. We let $\varepsilon(r)$ be the energy generated per unit mass within the shell at radius r. The increase in luminosity in that layer, due to energy generation, is then

$$dL = 4\pi r^2 \rho(r)\varepsilon(r)\, dr \tag{9.54}$$

The rate at which the luminosity changes with radius is then

$$dL/dr = 4\pi r^2 \rho(r)\varepsilon(r) \tag{9.55}$$

To carry out model calculations we must be able to specify $\varepsilon(r)$ as a function of composition and temperature. This is where the input from nuclear physics is important.

If, at any r, the radiative temperature gradient becomes too large, then convection will set in. The quantitative effect of convection is discussed in Section 23.3.2. Convection is an adiabatic process, no energy is lost from parcels of material as they move outward. At that point the temperature gradient becomes the one appropriate for adiabatic processes. This is called the adiabatic lapse rate.

We might wonder why stars don't explode with all the energy they produce. The answer is that their stability comes from their negative heat capacity. (Heat capacity is the energy required to raise the temperature by a given amount.) To see this, we look at the total energy $E = U + K$ (where K is now the kinetic energy of the thermal motions of particles in the gas). In Chapter 13, we will derive a relationship between total, kinetic and potential energies for bound systems, called the virial theorem, which tells us that $E = -K$. Suppose we now add energy. This increases E, making it less negative. This makes K less positive, so the total thermal kinetic energy decreases. The gas cools. Therefore, if the star produces too much energy for its equilibrium configuration, it can expand and cool to adjust. The above argument doesn't apply to degenerate gases, since their thermal energy is essentially independent of the temperature. Therefore, as we encounter degenerate stars, or parts of stars, we will see that explosions are possible.

9.5 Stellar models

In the previous section we saw a group of equations that describe the physics that governs stellar structure: (1) mass continuity, (2) hydrostatic equilibrium, (3) equation of state, (4) energy transport, and (5) energy generation.

The inputs to stellar models are the mass of the star and the composition. We must also specify the nuclear physics, which gives the energy generation as a function of these conditions. We must also put in the information about the opacities. For the purposes of calculation, we break the star into spherical shells. We must solve for the distribution of density $\rho(r)$ and temperature $T(r)$ which satisfy the conditions imposed by the equations of stellar structure. This is such a complicated process that realistic stellar models are all calculated on computers.

Stellar model calculations can also be used to predict stellar evolution. As nuclear reactions take place in the hot central core of the star, the composition changes. This changes the energy generation rate and the opacity, meaning the structure will change. We do a model calculation for the initial composition. We then determine the rate at which various nuclei are produced or destroyed. We then know what the composition will be some time, say 50 000 years, later. We now calculate a model with the altered composition, and then repeat the process. We follow the evolution of the star in these time steps. We choose the time steps so that the composition changes somewhat, but not too much, during each time step.

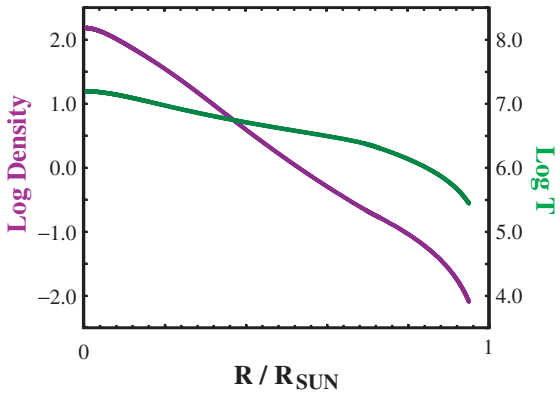

Fig 9.11 Temperature and density as a function of distance from the center of the Sun, as calculated from the solar model which best agrees with the global properties (radius, surface temperature) of the Sun.

To see the results of a model calculation, we look at a model for the Sun. The distribution of temperature and density is shown in Fig. 9.11.

9.6 Solar neutrinos

Though our model for the Sun gives the correct radius and temperature, there are certain aspects we cannot check directly. Almost all of the direct information we receive from the Sun comes from photons emitted in the solar atmosphere. We cannot directly observe photons that are emitted in the nuclear reactions in the center. Those photons are quickly absorbed and their energy takes about 10^7 years to reach the surface. We have no direct observations of the solar core now.

There is one opportunity to make a direct observation of the solar core. Neutrinos created in nuclear reactions in the core escape at the speed of light, virtually unattenuated. They reach us 8.5 min after they are created. We could use our stellar models to predict the rate at which neutrinos are emitted, and then try to measure the flux of neutrinos at the Earth. This would be a direct test of the solar model. The problem is that neutrinos are very hard to detect. If the whole Sun cannot absorb many neutrinos, a detector on Earth will absorb even fewer. The neutrino produced in the first reaction of the proton–proton chain ($p + p \rightarrow d + e^+ + \nu$) has a relatively low energy, and this makes it particularly hard to detect.

There is, however, a source of higher energy neutrinos. Once ^{3}He is formed, most of it reacts to form ^{4}He, as discussed in Section 9.2. However, in a small fraction of cases, the following reaction can take place:

$$^3\text{He} + {}^4\text{He} \rightarrow {}^7\text{Be} + \gamma \tag{9.56}$$

The ^{7}Be then captures a proton:

$$^7\text{Be} + \text{p} \rightarrow {}^8\text{B} + \gamma \tag{9.57}$$

The boron then beta decays, emitting a neutrino in the process:

$$^8\text{B} \rightarrow {}^8\text{Be} + \text{e}^+ + \nu \tag{9.58}$$

The ^{8}Be then breaks apart into two alpha particles:

$$^8\text{Be} \rightarrow {}^4\text{He} + {}^4\text{He} \tag{9.59}$$

The neutrino emitted in the beta decay of the ^{8}B has enough energy to provide us with some chance of detecting it.

We can detect this neutrino using an isotope of chlorine, ^{37}Cl. About 25% of all naturally occurring chlorine is in the form of this isotope. When struck with a sufficiently high energy neutrino, it can absorb the neutrino:

$$^{37}\text{Cl} + \nu \rightarrow {}^{37}\text{Ar} + \text{e}^- \tag{9.60}$$

This particular isotope of argon is radioactive, and its decay can be detected in normal particle detectors in the laboratory. If we start with a tank of chlorine (and no radioactive argon), and we end up with a small amount of ^{37}Ar, we can determine the rate at which neutrinos were hitting the tank.

This is the basic idea behind an experiment conducted by R. Davis of the Brookhaven National Laboratory. The source of liquid chlorine is perchloroethylene (cleaning fluid). Since neutrinos interact infrequently, a large quantity of "detector" is used – about 10^5 gallons. The argon is produced as an inert gas, so it will form bubbles in the fluid, allowing it to be removed. The experiment is run for some period of time, typically a month. The gas is then collected, and is measured to gauge ^{37}Ar activity.

Even if some radioactive argon is found, we don't know that it results from solar neutrinos.

There are also high energy particles in the Earth's atmosphere, called cosmic rays, which can produce a similar result. The Earth can shield the tank from cosmic rays but not from neutrinos. The tank is therefore placed 1.5 km underground in the Homestake gold mine, near Lead, South Dakota. Another source of possible contamination is natural radioactivity from the rocks in the mine. For this reason, the tank is surrounded by a larger tank containing water. The water blocks the high energy particles from radioactive decays in the mine, but doesn't block neutrinos. You can see from this brief description that this experiment is a very difficult one.

The results of the experiment from some 20 years observing have been astounding. It is convenient to express the rate of detections in *solar neutrino units* or *SNU*. The standard model of the Sun predicts that we should be measuring 8 SNU. The experiment, which detects about one event every two days, yields 2.6 SNU, a number that is only about one-third of this value. This leads to what we call *the solar neutrino problem*. Astrophysicists have so much faith in our understanding of stellar interiors that such a large discrepancy is an indication of a severe problem.

A number of solutions have been suggested. One possibility is that there is something wrong with the experiment itself. Possibly, the detector is not as sensitive to neutrinos as originally thought. However, various aspects of the experiment have been checked and refined over the years, and there is a general feeling that the experiment is correct.

If the experiment is correct, then the solar model must be examined. It is possible that some of the inputs are not correct. For example, the nuclear reaction rates have a very strong temperature dependence. If we have that dependence slightly wrong, then significant changes can occur in the solar model. It is also possible that the opacities (as a function of composition and temperature) are slightly off. The solar neutrino problem has stimulated work in these areas. We now have improved nuclear physics data and opacity data. A better solar model has been calculated, with a much smaller uncertainty, but the predicted neutrino flux remains essentially the same. There is an additional check on the solar model. It comes from helioseismology (discussed in Chapter 6), in which the oscillations of the Sun are studied. The solar model (temperature, density and composition vs. radius) can be used to predict the details of these oscillations. The agreement between theory and experiment is very good, meaning that the solar model is probably not the cause of the solar neutrino problem.

It has been suggested that the Sun may go through a cycle in its energy generation, and that right now it is generating less than the average amount of energy. In this cycle, at some point the core cools, reducing the rate of nuclear reactions. The pressure decreases and the core contracts. As the core contracts, it converts gravitational potential energy into kinetic energy, and begins to heat. As the core becomes hotter the rate of nuclear reactions increases. The pressure increases and the core expands. The cycle then starts again. If this is the answer, then the neutrino observations are giving us a good view of what is happening in the Sun now. In this picture, it is possible for the cycle to take place with very little variation in the solar luminosity. Since photons are scattered many times before they can go from the center to the photosphere, the light from the photosphere reflects the average energy production over the Kelvin time. It should be pointed out that no mechanism for such oscillations has been found.

It has been suggested that we don't know as much as we thought we did about neutrino physics. We now know that there are three different neutrino types, each with its own antineutrino. (We'll discuss particle physics more in Chapter 21.) The type of neutrino that we have been discussing is called the electron neutrino, because it always appears in reactions with electrons (or positrons). This is the type of neutrino produced in the Sun, and the type that will interact with ^{37}Cl. However, some theories have suggested that neutrinos can change their identities. If this is the case, then a neutrino can be created as an electron neutrino in the Sun, but change its identity by the time it reaches the Earth. According to this idea, as few as one-third of the neutrinos produced in the Sun might be capable of being absorbed by the ^{37}Cl. These identity changes, or *neutrino oscillations*, are also related to the suggestions that neutrinos have a very small (but not

zero) mass. The experimental evidence for this is being studied, and new experiments are underway.

In the meantime two new types of solar neutrino experiments are being done. The first involves gallium as a detector. Since the ^{8}B neutrino comes from a relatively unimportant branch in the Sun's nuclear reaction chain, it may be that there is a small error in our calculations that just happens to be magnified for this minor branch. There is another reaction that also produces a detectable neutrino. The reaction is not as important as the proton–proton chain, but it is more significant than the branch involving ^{8}B. This reaction is called the *p–e–p reaction*:

$$p + e^- + p \rightarrow d + \nu \tag{9.61}$$

It occurs once for every 400 direct p–p reactions. It is rare because it is much harder to bring three particles together at the same time than it is to bring two particles together.

The neutrino produced by the p–e–p reaction is not as high in energy as the ^{8}B decay neutrino, and is not absorbed by ^{37}Cl. However, it has a higher energy than the p–p neutrino. The p–e–p neutrino can be absorbed by gallium. As with chlorine, large quantities of gallium are needed for the experiments. Two such experiments are being carried out, a European–Israeli–USA collaboration known as GALLEX (30 tons of gallium), and a Russian–USA collaboration known as SAFE (60 tons of gallium). These experiments yield about one event per day. The measured rates are about half of the predicted values.

The second set of new experiments involves larger versions of traditional detectors used in elementary particle physics experiments. These involve large amounts of water as detectors, with each reaction producing a small flash of light, and that light is detected by an array of photomultipliers. The first set of experiments was carried out by a Japanese group at Kamiokande. The experiment was originally set up as a particle physics experiment, but it turned out that it was also capable of detecting solar neutrinos. An important feature of this detector is that it can measure the direction from which the neutrinos are arriving, and the group were able to verify that the neutrinos are, indeed, coming from the Sun (and not, for example, from some background contamination). Following their original success, they built a second version, known as SuperKamiokande, more closely designed for observing solar neutrinos. The results of these observations is that observed flux is about half the theoretical predictions. For their work on this problem, Davis and M. Koshiba, who headed the Japanese project, shared the 2002 Nobel Prize in Physics.

At this point, it appears that the best solution to the solar neutrino problem involves the neutrinos. If they have a very small mass, and can change identities, then the theory and experiment can be brought into agreement. In 2002, a group in Sudbury, Canada (Sudbury Neutrino Observatory, SNO, shown in Fig. 9.12), provided strong evidence that neutrino oscillations are the answer. An analysis of their results

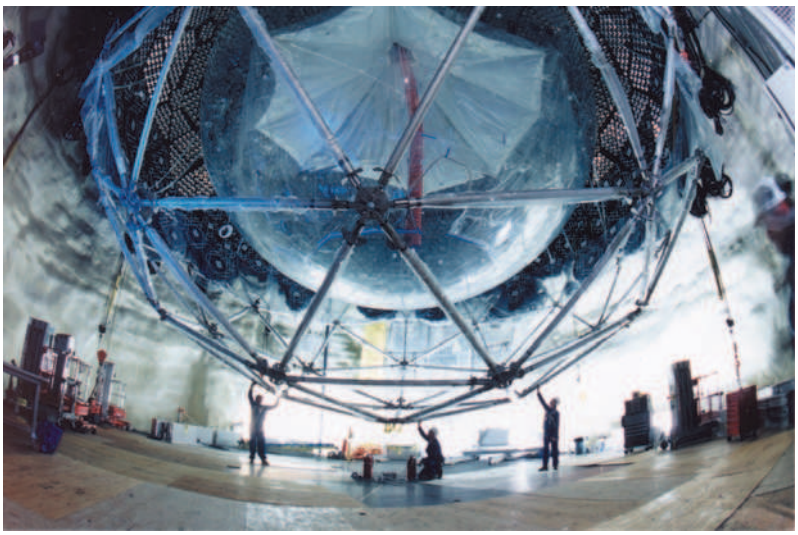

Fig 9.12 The large detector of the Sudbury (Canada) Neutrino Observatory (SNO). [Photo courtesy of Sudbury Neutrino Observatory]

suggests that the neutrino mass is in the 10^{-2} eV range. (Remember, for comparison, the mass of the proton is almost 10^9 eV.) The analysis also shows that there is a significant chance (of the order of 50% with a large uncertainty) for neutrinos created in the Sun to change their identity by the time they reach our detectors. The next major test of these ideas would be an experiment to detect directly the p–p neutrinos (by far the most abundant, and lowest energy).

Chapter summary

In this chapter we looked at the processes responsible for the structure of main sequence stars.

We started by looking at energy sources. Nuclear reactions are the only source capable of giving stars their inferred lifetimes. We saw that the temperatures required for nuclear reactions to take place are in excess of 10^7 K, even with tunneling to help bring the nuclei together.

The basic source of energy on the main sequence is the conversion of hydrogen to helium. In low mass stars this takes place primarily via the proton–proton chain. In more massive stars, with higher central temperatures, other cycles, such as the CNO cycle, are important.

We also looked at the basic processes that govern stellar structure. We saw that normal stars are in hydrostatic equilibrium, with each layer supported by the pressure difference between its bottom and top. We also saw that the temperature distribution is determined by the requirement that the temperature of each layer be constant.

Once the basic laws of stellar structure are outlined, stellar models can be computed, generally using computers. In a model, we start with a certain mass and composition, and calculate the equilibrium configuration.

We saw how much of stellar structure seems to be understood, but encountered the puzzle of the solar neutrino experiment. The neutrinos allow us to see what the core of the Sun is doing now, and it does not appear to be doing as much as models predict. The answer to this problem seems to lie in neutrons changing their identity in their trip from the Sun.

Questions

9.1. (a) Why is 1 eV/atom a reasonable estimate for the energy available in chemical reactions? (b) Is the estimate likely to be wrong by more than a factor of ten in either direction? Explain. (c) If the estimate is wrong by a factor of ten in either direction, will it change the conclusion that the Sun cannot exist on chemical reactions?

9.2. It has been said that if we did not know that $E = mc^2$, then we would not know about nuclear energy. Discuss this.

9.3. Explain the factors that place upper and lower limits on the number of neutrons that go into a nucleus with some specific number of protons.

9.4. Why can't a free proton beta decay into a neutron?

9.5. What are the similarities between gamma emission by nuclei and visible light emission by atoms?

9.6. Why is there no Coulomb barrier to fission if there is one to fusion?

9.7. Why are the rates of certain fusion reactions very sensitive to temperature?

9.8. Why are the r- and s-processes important?

9.9. (a) What are the parameters that we put into a stellar model? (b) What calculations do we perform? (c) How do we test the results?

9.10. (a) What do we mean by spherical symmetry? (b) Why will a rotating star not be spherically symmetric?

*9.11 In equation (9.26) we have used the form of the gravitational force between two point particles. These particles have masses $M(r)$ and dm and are a distance r apart. However, $M(r)$ represents an extended mass and dm represents a shell, so neither represents a point. How can we use the simple formula? (Hint: Treat the justification for $M(r)$ and dm separately.)

*9.12 What effect would a slight increase in opacity at all layers have on the structure of a star?

9.13 Explain how we simulate stellar evolution.

9.14 Why is the solar neutrino problem so important?

9.15 When we considered explanations of the solar neutrino experiment, we said that the Sun may be generating some energy now through gravitational collapse. However, earlier in the chapter, we ruled out gravitational collapse as a stellar energy source. Why isn't this a contradiction?

*9.16 The equilibrium structure of a star is ultimately determined by its mass and composition. Show that the structure of a star determines the rate of energy generation and not the other way around.

Problems

9.1. What is the gravitational potential energy of (a) the Sun, (b) a 1 $M_\odot$ black hole?

9.2. (a) What is the gravitational potential energy of an interstellar cloud, with a density of 1000 H atoms/cm^3 and radius of 10 pc? (b) What is its kinetic energy if its temperature is 10 K?

*9.3. Find the gravitational potential energy of a sphere with a $1/r^2$ density distribution. Take the total mass of the sphere to be M, and let the density $\rho(r) = \rho_0/r^2$ out to a radius, R. Express your final answer in terms of M and R.

9.4. Estimate the lifetime of a 10 $M_\odot$ star on the main sequence to give off energy stored from gravitational collapse.

9.5. Calculate the mass corresponding to the binding energy of an H atom. What fraction of the mass of the atom is this?

9.6. What is the rate at which the Sun is converting mass to energy?

9.7. (a) What is the difference in mass between the neutron and the proton, expressed in MeV? (b) How does this relate to the energy available in the beta decay of a neutron?

9.8. Calculate the binding energy of a ^{4}He nucleus.

9.9. How much energy per proton is given off in the p–p chain? (Express your answer in MeV.)

*9.10. Suppose we have Z protons and have to distribute them into two nuclei, one with Z_1 protons and the other with Z_2 protons ($Z = Z_1 + Z_2$). (a) What arrangements give the maximum and minimum Coulomb repulsion between the two nuclei? (b) What does this tell you about the types of fusion that are most likely to take place in stars?

9.11. (a) How close can two protons get if one is at rest and the other has a kinetic energy equal to the average energy at $T = 10^7$ K? (b) What is the wavelength of the moving proton, and how does it compare with the minimum separation between the two protons? (c) Repeat the calculations for a proton with ten times the average energy at this temperature.

*9.12. Suppose the density of a star is given by

$$\rho(r) = \begin{cases} \rho_0 & r < r_0 \\ \rho_0(r_0/r)^2 & r_0 < r < R \\ 0 & R < r \end{cases}$$

(a) Find an expression for $M(r)$. (b) If the mass of the star is 1 $M_\odot$ and $R = R_\odot$, and $r_0 = 0.1\ R_\odot$, what is the value of ρ_0?

9.13. For the density distribution in the previous problem, find $P(r)$.

Computer problems

9.1. Calculate the gravitational potential energies for stars in the middle of each spectral class on the main sequence (O5, B5, . . .).

9.2. Write a routine that will calculate the gravitational potential energy for uniform density elliptical objects which have two axes the same. Evaluate the potential energy for the two following cases, both clouds with 100$M_\odot$ of material: (a) an oblate (flattened) ellipsoid with semi-axes 10, 10 and 5 pc; (b) a prolate (elongated) ellipsoid with semi-axes 5, 5, and 10 pc.

9.3. Estimate the nuclear reaction lifetime of a star as a function of its mass. Assume that 10% of the mass is in the core and available for nuclear reactions. (Hint: Use the mass–luminosity relationship.) Plot your results for the range of masses encountered on the main sequence.

9.4. Make a table showing the Kelvin time for stars in the mid-range of each spectral type (O5, B5, . . .).

9.5. Make a graph of the electrical potential energy between two protons as a function of their separation, r. Let r range from the size of a nucleus to ten times that.